

Mini Project

Abstract

Title

MediLock – Multi-Hospital Digital Health Record & Care Coordination Platform

Abstract

MediLock is a secure web-based healthcare platform designed to connect multiple participating hospitals through a unified digital health record system. It provides patients with a lifelong digital health locker while enabling authorized doctors from registered hospitals to securely access a patient's medical history after obtaining patient consent through OTP or QR-based verification. The platform addresses the common problem of fragmented medical records when patients visit different hospitals, reducing duplicate diagnostic tests, improving treatment decisions, and ensuring continuity of care. The system emphasizes privacy, secure access control, and seamless collaboration among participating hospitals while allowing patients to maintain complete ownership of their medical records.

Technology Stack

- Frontend: React.js
- State Management: Redux Toolkit
- Backend: ASP.NET Core Web API
- Database: Microsoft SQL Server
- ORM: Entity Framework Core
- Authentication: JWT Authentication
- Security: Role-Based Access Control (RBAC)
- OTP Service: Twilio/SMS or Email OTP

User Roles

1. Super Admin

- Approve/reject hospital registrations
- Manage all hospitals
- Manage all users
- Assign roles
- View audit logs
- View system analytics
- Disable accounts

Dashboard includes:

- Total Hospitals
- Total Patients
- Total Doctors
- Total Medical Records
- Active Users

2. Hospital Admin

- Register/Login
- Manage hospital profile
- Add/Edit/Delete doctors
- Add/Edit/Delete lab technicians
- Manage departments
- View appointments
- View hospital reports

Cannot:

- Access another hospital's staff
- Access patient records without patient consent

3. Doctor

- View appointments
- View patient history (after OTP/QR approval)
- Add diagnosis
- Create prescriptions
- Upload medical records
- Request access to previous records
- View lab reports

Cannot:

- Delete patient history
- Access records without permission

4. Lab Technician

- View assigned lab requests
- Upload test results
- Upload medical images
- Mark tests as completed

Cannot:

- Edit prescriptions
- View unrelated patient information

5. Patient

- Register/Login
- Update profile
- Book appointments
- View prescriptions
- Download reports
- Share records using OTP or QR
- View access history
- Revoke record access

Mini Project Modules

- Patient Registration and Secure Login
- Hospital Registration and Approval
- Hospital Administrator Dashboard
- Doctor Management
- Appointment Booking and Scheduling
- Digital Prescription Management
- Medical Record Upload (Prescriptions, Lab Reports, Discharge Summaries, Medical Images)
- OTP/QR-Based Record Sharing Between Participating Hospitals
- Role-Based Access Control (Super Admin, Hospital Admin, Doctor, Lab Technician, Patient)
- Audit Log for Medical Record Access

Major Project Extension

The major project extends the web application into a comprehensive digital healthcare ecosystem by introducing intelligent healthcare services, mobile accessibility, and advanced analytics.

- Patient Mobile Application
- AI-Based OCR for Paper Prescription Digitization
- Medicine Reminder and Follow-up Notifications
- Integrated Pharmacy Module
- Emergency QR Health Card
- Cross-Hospital Medical Record Sharing
- Health Analytics Dashboard
- Notification System (SMS/Email/Push)
- Hospital Performance Analytics

Example Scenario

1. Patient registers on your platform.
2. Patient visits **Hospital A**.
3. Doctor at Hospital A adds a diagnosis and prescription.
4. Two months later, the patient visits **Hospital B**, which is also on the platform.
5. With the patient's consent (OTP or QR code), the doctor at Hospital B can view the patient's previous records from Hospital A.
6. Hospital B adds new records, creating a continuous medical history.